

Linear Regression and Endogeneity

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What is Regression?

Economic relationship

- An economic model consists of mathematical equations that describe various relationships:

$$y = f(x_1, x_2, x_3, x_4, x_5)$$

Regression

- Regression is a statistical method that analyzes the relationship between a **dependent variable** (y) and one or more **independent variables** (x_1, x_2, x_3, x_4, x_5).

Problems When Doing A Regression

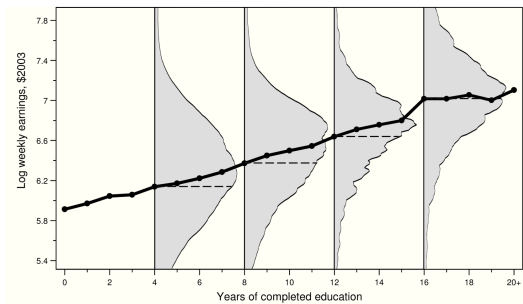
Problems when doing a regression:

- The form of the equation $f(\cdot)$ must be specified.
- Determine which factors to incorporate.
- Deal with the randomness.

Why randomness?

An Example

Suppose we want to study how years of schooling impact people's weekly wages.



- Earnings vary even if the sample has the same schooling.
- There's a positive relationship between schooling and expected earnings.
- Is it also positive for the whole population?

The Best Function Form

In the schooling and earnings example,

$$y = f(x_1, x_2, x_3)$$

- y is weekly earnings;
- x_1 is years of formal education;
- x_2 is years of previous working experience;
- x_3 is an indicator variable for training, equal to 1 if the individual received training before starting the job.

What form of f is the best to reveal the relationship between years of schooling and earnings?

- "Best" typically means the Minimum Mean Squared Error (MMSE):

$$\text{mean squared error} = E[(Y - f(x_1, x_2, x_3))^2]$$

Conditional Expectation Function

- The conditional expectation of Y given $X = x$ is defined as:

$$E[Y|X = x] = \sum_y y \cdot P(Y = y|X = x)$$

- Theorem 3.1.2 in *Mostly Harmless Econometrics*
 - ▶ The conditional expectation function (CEF) is the MMSE predictor of Y given X .
 - ▶ The CEF is the best predictor in the sense of MMSE:

$$y = f(x_1, x_2, x_3) = E[Y|X_1 = x_1, X_2 = x_2, X_3 = x_3]$$

Linear Regression

We assume the conditional expectation function (CEF) to be a linear function:

$$E[Y|X = x] = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3,$$

where β solves

$$\min_{\mathbf{b}} \{E[(Y - (b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3))^2]\}$$

The advantages of assuming a linear form:

- Simplicity
- Interpretability
- ...

Validity of Linear Form

Is it sufficient to use the linear form?

- If the assumption is true, i.e., if CEF is linear, then the population regression function is it.
- $\sum_i \beta_i X_i$ is the best linear predictor in a MMSE sense.
- $\sum_i \beta_i X_i$ provides the MMSE linear approximation to $E[Y|X]$.

The above statements are Theorem 3.1.4 to Theorem 3.1.6 in *Mostly Harmless Econometrics*.

Recap

Now you probably have understood:

- What is regression?
- What's the best predictor?
- Why do we use linear regression?

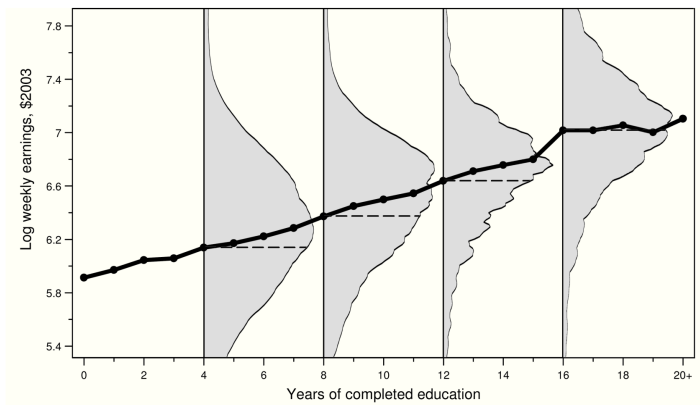
Recap

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- What's the best predictor?
- Why do we use linear regression?

Then we will have a practice!

The Schooling Example

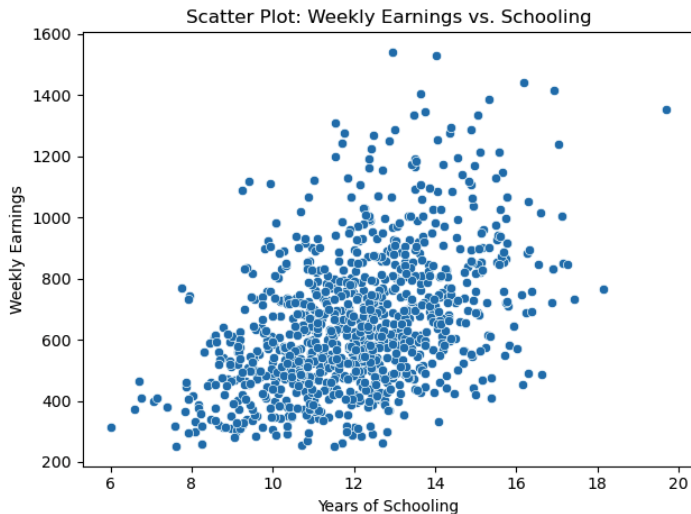


- The observed facts (or intuition): Years of schooling are positively correlated with weekly earnings.
- Research Question: Does more schooling **lead to** higher earnings?

Step 1 and Step 2: Question and Hypotheses

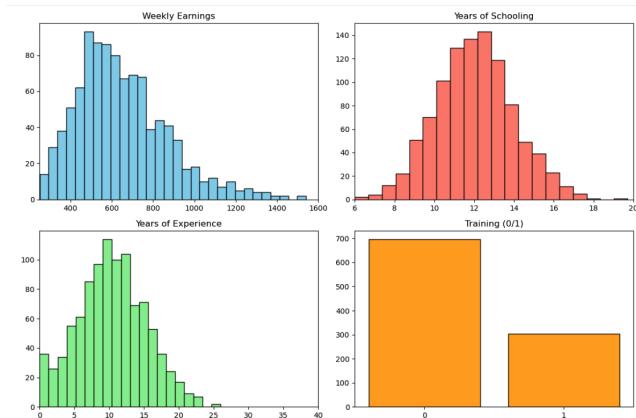
- Step 1: Clarify the research question.
 - ▶ Research Question: Does more schooling **lead to** higher earnings?
 - ▶ Y: Earnings
 - ▶ X: Years of schooling
- Step 2: Form the Hypotheses.
 - ▶ H0: Schooling does not impact earnings.
 - ▶ H1: Schooling positively impacts earnings.

Step 3: Observe Your Data



Step 3: Observe Your Data Cont.

Summary Statistics					
	Mean	Variance	Median	Min	Max
Weekly_earnings (\$)	651.76	49579.04	615.48	250.58	1541.49
Schooling (years)	12.04	3.83	12.05	6	19.71
Previous Working Experience (years)	10.4	23.78	10.32	0	25.97
Training Indicator	0.3	0.21	0	0	1



Step 4: Form the Regression Function

In the schooling and earnings example,

$$Y = b_0 + \beta X + \beta_2 X_2 + \beta_3 X_3 + \epsilon$$

- Y is the **dependent variable**: weekly earnings;
- X is the **independent variable**: years of schooling;
- X_2 and X_3 are control variables:
 - ▶ X_2 is years of previous working experience;
 - ▶ X_3 is an indicator variable for training, equal to 1 if the individual received training before starting the job.
 - ▶ ...
- ϵ is the error term.
- β is the coefficient of interest.

Step 4: Form the Regression Function Cont.

How do we determine the control variables?

- Control all the **confounders**: variables that affect both X and Y .
- Do not control post-treatment variables (mediators).
- Do not control irrelevant variables.
- Be cautious about collinearity.

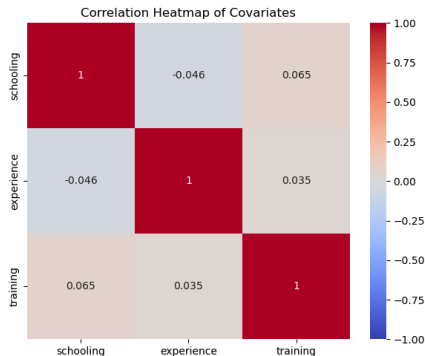
How do we know in practice?

- Economic theory / subject knowledge
- Common sense
- Literature

Check the Collinearity

There are several ways to check the collinearity:

- Covariate Correlation Matrix
- Variance Inflation Factor (VIF)
- Common sense



VIF Values	
Variable	VIF
Schooling (years)	1
Previous Working Experience (years)	1
Training Indicator	1

Step 5: Run the Regression

The Ordinary Least Squares (OLS) is the way to estimate β by minimizing the sum of squared errors (SSE):

$$\hat{\beta} = \arg \min_{\beta} \sum_{i=1}^n (y_i - \mathbf{x}_i^{\top} \beta)^2$$

```
import statsmodels.api as sm

# Define dependent variable
y = schooling["weekly_earnings"]

# Define independent variables (main regressor + controls)
X = schooling[["schooling", "experience", "training"]]

# Add intercept term
X = sm.add_constant(X)

# Fit the OLS model
model = sm.OLS(y, X).fit()

# Display the regression results
print(model.summary())
```

Step 5: Run the Regression Cont.

OLS Regression Results

```
=====
Dep. Variable:      weekly_earnings      R-squared:              0.599
Model:              OLS                  Adj. R-squared:         0.597
Method:             Least Squares        F-statistic:           495.1
Date:               Thu, 21 Aug 2025     Prob (F-statistic):    7.31e-197
Time:               15:16:03             Log-Likelihood:        -6367.7
No. Observations:   1000                 AIC:                   1.274e+04
Df Residuals:       996                  BIC:                   1.276e+04
Df Model:           3
Covariance Type:    nonrobust
=====
```

```
=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
const      -294.9920     29.891     -9.869     0.000    -353.650    -236.335
schooling    52.9745      2.292     23.115     0.000     48.477     57.472
experience   26.9123      0.918     29.309     0.000     25.110     28.714
training    96.0868      9.750      9.855     0.000     76.954    115.219
=====
```

```
=====
Omnibus:              92.706      Durbin-Watson:           2.074
Prob(Omnibus):        0.000      Jarque-Bera (JB):        133.486
Skew:                 0.701      Prob(JB):                1.03e-29
Kurtosis:             4.113      Cond. No.                 109.
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Step 5: Run the Regression Cont.

We really appreciate it if you display your regression results properly in your group projects. Example:

Regression Results	
Dep. Var: Weekly Earnings	OLS
Schooling	52.9745*** (2.292)
Experience	26.9123*** (0.918)
Training	96.0868*** (9.75)
Observations	1000
R-squared	0.599

Notes: This table reports results from an OLS regression of weekly earnings on schooling, controlling for experience and training.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Step 6: Interpretation

1 OLS Regression Results					5		
Dep. Variable:	weekly_earnings		R-squared:	0.599			
Model:	OLS		Adj. R-squared:	0.597			
Method:	Least Squares		F-statistic:	495.1			
Date:	Thu, 21 Aug 2025		Prob (F-statistic):	7.31e-197			
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Df Model:	3						
Covariance Type:	nonrobust		3	4			
	coef	std err	t	P> t	[0.025	0.975]	
const	2	-294.9920	29.891	-9.869	0.000	-353.650	-236.335
schooling		52.9745	2.292	23.115	0.000	48.477	57.472
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Kurtosis:		4.113	Cond. No.			109.	

Notes:

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The result shows:

$$\text{Weekly earnings} = \hat{\beta}_0 + \hat{\beta}_{OLS} \text{ Schooling} + \hat{\beta}_2 \text{ Experience} + \hat{\beta}_3 \text{ Training} + \epsilon$$

where $\hat{\beta}_0 = -294.99$, $\hat{\beta}_{OLS} = 52.97$, $\hat{\beta}_2 = 26.91$, and $\hat{\beta}_3 = 96.09$

Step 6: Interpretation Cont.

$$\text{Weekly } \hat{\text{earnings}} = -294.99 + 52.97 \text{ Schooling} + 26.91 \text{ Experience} + 96.09 \text{ Training}$$

Key points:

- Schooling has positive impacts on weekly earnings.
- $\Delta(\text{Weekly earnings}) = 52.97\Delta(\text{Schooling})$
- Increasing one-year schooling will lead to an increase of \$52.99 per week.

Step 6: Interpretation Cont.

$$\text{Weekly } \hat{\text{earnings}} = -294.99 + 52.97 \text{ Schooling} + 26.91 \text{ Experience} + 96.09 \text{ Training}$$

Key points:

- Schooling has positive impacts on weekly earnings.
- $\Delta(\text{Weekly earnings}) = 52.97\Delta(\text{Schooling})$
- Increasing one-year schooling will lead to an increase of \$52.99 per week.

Now $\hat{\beta}_{OLS} = 52.97$ is economically significant (far from 0), but we still need to test if it's significantly different from 0 statistically (Why?).

Step 7: Test the Hypotheses

The hypotheses we have:

- H_0 : Schooling does not impact earnings.
- H_1 : Schooling positively impacts earnings.

More precisely,

- $H_0: \hat{\beta}_{OLS} = \beta = 0$
- $H_1: \hat{\beta}_{OLS} > 0$

Test the hypotheses:

- Determine the significance level α .
- Determine if it is a two-sided test or a one-sided test.

Background Knowledge

What is a significant level?

- The significance level α is the probability of rejecting the null hypothesis when it is actually true. Usually we set it at 5% or 1% level, i.e., $\alpha = 0.05$ or 0.01 .

What are two-sided tests and one-sided tests?

- Two-sided test: used when we only care whether a coefficient is different from zero.
- One-sided test: used when we have a directional hypothesis (e.g., we believe education raises earnings, not lowers them).

Step 7: test The Hypotheses

Suppose we pick the $\alpha = 0.05$.

Construct the test statistic:

$$t = \frac{\hat{\beta}_{OLS} - \beta}{SE(\hat{\beta}_{OLS})}$$

- $\hat{\beta}_{OLS}$ is the estimation.
- β is the true value of the whole population.
- $SE(\hat{\beta}_{OLS})$: standard error of the estimate.

Then according to the regression result, the calculation of t is

$$t = \frac{52.97 - 0}{2.292} = 23.111$$

t-test Table

t Table

cum. prob one-tail	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.282	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
Confidence Level											

Step 7: Test the Hypotheses Cont.

$$t = \frac{52.97 - 0}{2.292} = 23.111$$

And by checking the t-table:

- If we do a one-sided test: $t_{0.05} = 1.645$
- If we do a two-sided test: $t_{0.05} = 1.960$
- $t > t_{0.05}!!$

The good news implies **there is enough evidence to support us to reject H_0** . Hence, $\hat{\beta}_{OLS}$ is significantly different from 0. And now we can conclude that education positively impacts earnings economically and statistically.

Recap

- Clarify the research question
- Form hypotheses
- Check the data
- Determine the control variables
- Run the regression
- Interpretation
- Test the significance of the coefficient of interest
- Make a conclusion

Evaluate the Estimator

Is it a **good** estimator? To what extent can we trust and rely on the estimator?

There are two dimensions to measure the estimator:

- Does it truly reveal the impacts on the whole population?
- Does it reflect the causal relationship?

From Sample to the Whole Population

To make the estimator truly reveal the coefficient of interest of the whole sample, we need to check if the estimator is:

- Unbiased: $E[\hat{\beta}_{OLS}] = \beta$.
- Consistency: $\hat{\beta}_{OLS}$ converges in probability to β as $n \rightarrow \infty$.
- Efficiency: Among all **linear unbiased estimators**, $\hat{\beta}_{OLS}$ has the smallest variance.

From Sample to the Whole Population Cont.

$\hat{\beta}_{OLS}$ has the above properties if the following assumptions are true (by Gauss-Markov Theorem):

- **(A1)** Random sampling: the data (Y_i, X_i) are independently and identically distributed.
- **(A2)** No perfect Multicollinearity: $X^T X$ is non-singular.
- **(A3) Zero Conditional Mean of Errors (Exogeneity):** $E[\epsilon|X] = 0$ (for unbiasedness and consistency).
- **(A4)** Homoskedasticity: $Var(\epsilon|X) = \sigma^2 I$ (for efficiency).

Causal Relationship

What is a causal relationship?

- It tells how a change in X causes a change in Y .

When can we say $\hat{\beta}_{OLS}$ reflects the causal relationship between earnings and schooling?

- Theoretically, if we are able to prove (A3) holds (not an assumption anymore), then $\hat{\beta}_{OLS}$ can be interpreted as revealing a causal relationship.

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Do we have a credible reason to believe that (A3) holds in our context?

What Does Exogeneity Mean?

(A3) Zero Conditional Mean of Errors (Exogeneity): $E[\epsilon|X] = 0$ (for unbiasedness and consistency).

- $E[Y - \beta_0 - \beta_1 X - \beta_2 X_2 - \beta_3 X_3 | X] = 0$.
- it means everything we do not observe is unrelated to X and has a zero mean.

Practically, it means:

- We have included all variables affecting both X and Y .
- There is no reverse causality, i.e. Y does not impact X .
- There is no measurement error.

Does Exogeneity Hold?

The regression function is

$$Y = \beta_0 + \beta X + \beta_2 X_2 + \beta_3 X_3 + \epsilon$$

- Y is the **dependent variable**: weekly earnings;
- X is the **independent variable**: years of schooling;
- X_2 and X_3 are control variables:
 - ▶ X_2 is years of previous working experience;
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Does Exogeneity Hold? Cont.

The biggest problem with our model is that we have too many omitted variables:

- Innate ability
- Family background
- Motivation
- Access to resources
- ...

There are too many confounders in the error term, so $\hat{\beta}_{OLS}$ captures not only the impact from X to Y , but also the impact of the confounders to Y .

In conclusion, $\hat{\beta}_{OLS}$ is not causal.

How Can We Ensure the Exogeneity?

By research design or theory:

- Random assignment
- A valid instrumental variable
- A convincing natural experiment